Phase 1

Network Security Design & Implementation

Project Information

Course Name	Network Security
Course Number	0603443
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Semester	First Semester
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1. Project Overview & Objectives

1.1 Project Scope

Our project focuses on designing a secure enterprise network topology and applying modern network security principles to protect it from common threats. The work will include implementing security mechanisms such as firewalls, DMZ, intrusion detection systems, access control techniques, explaining practically the principles of honeypot and SIEM.

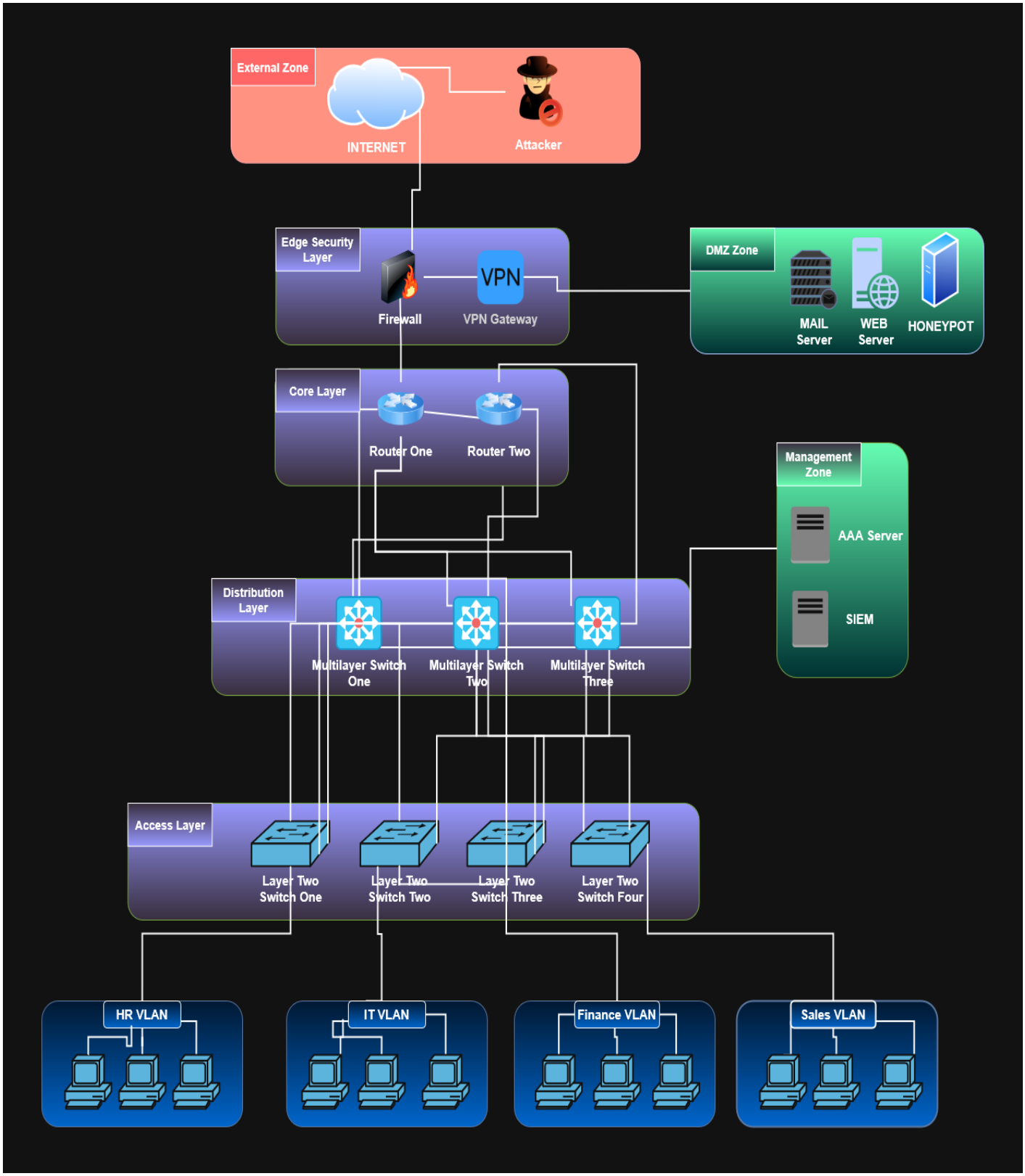
1.2 Objectives

The main objective of our project is to design and secure a network that simulates a real enterprise environment.

The project aims to understand how different security components work together to protect a network, such as firewalls, intrusion detection systems, and secure zones like DMZ.

In addition, the project will focus on monitoring network activity and analyzing traffic to detect any suspicious behavior

2. System Description & Topology



The proposed network will follow a three-tier architecture, including core, distribution, and access layers.

A DMZ zone will be created to host public services such as web servers, separated from the internal network.

Security mechanisms such as firewalls, IDS/IPS, and AAA will be integrated to enhance protection and control access.

Following is a brief description for the components of the network:

2.1 Edge Security Layer

Firewall

The firewall is the first line of defense between our network and the internet. It controls all incoming and outgoing traffic based on predefined rules.

VPN Gateway

The VPN gateway allows remote users or branch offices to securely connect to the network.

2.2 Core Layer

The core layer contains two routers working together to ensure the network is always available.

If one router fails, the other immediately takes over, so there is no downtime. This setup removes a single point of failure, that is an example of fault tolerance.

2.3 Distribution Layer

The distribution layer includes three multilayer switches that connect the core layer to the access layer. They handle routing between VLANs, apply access control rules.

2.4 Access Layer

The access layer connects end-user devices like PCs and laptops to the network. Switches in this layer are configured with VLANs to separate departments.

2.5 DMZ Zone

The DMZ is a separate part of the network used to host public services, inclues

- Mail Server: handles inbound and outbound email traffic
- Web Server: hosts the organisation's web-facing services.
- Honeypot: fake system used to attract attackers.

2.6 Management Zone

AAA Server

The AAA server manages authentication for users and network devices. It ensures that only authorized users can access the system, and it records all activities for monitoring.

SIEM

The SIEM system collects logs from all network devices and security tools, then it analyzes these logs to detect suspicious activities or attacks that may not be obvious from a single device.

2.7 VLAN Segmentation

The network is divided into VLANs based on departments to improve security and organization, includes

- HR VLAN
- IT VLAN
- Finance VLAN
- Sales VLAN

3. Expected Tools

In this project, GNS3 will be used as the main platform to design and simulate the network topology, allowing more realistic configurations and advanced features.

Security mechanisms such as firewalls will be implemented through router configurations and access control lists (ACLs).

For intrusion detection and prevention, Snort will be configured to operate in both IDS and IPS modes.

Splunk will be used as a SIEM system to collect and analyze logs from different network components.

Additionally, tools like Nmap and Wireshark will be used for network scanning and traffic analysis, while Metasploit will be used to simulate attack scenarios and test the effectiveness of the implemented security measures